

CLEVIS User Manual

Command Line Vector Graphics Software

Group: 18

Group members: Meng Zihan, Huang Junhao, Luo Yeyushan, He Liuying

1. Introduction

Welcome to **CLEVIS** (Command Line Vector Graphics Software) created by group 18. This tool allows you to create, manipulate, and organize vector graphics entirely through a command-line interface. It is designed for users who require precise control over geometric shapes and their arrangements.

Key Features:

- **Precision Drawing:** Create Rectangles, Squares, Circles, and Lines using exact coordinates.
- **Grouping:** Organize multiple shapes into a single logical unit (Group) for easier manipulation.
- **Z-Order Management:** Automatically handles the layering of shapes (newer shapes appear "on top").
- **History Management:** Robust Undo and Redo capabilities to correct mistakes.
- **Automated Logging:** All operations are automatically recorded to HTML and TXT log files for future reference.
- **Shape Creation:** Every shape must have a unique name.
- **Shape Naming Rules:** Every shape must have a unique name. Names must consist **only** of letters (a-z, A-Z), numbers (0-9), and underscores (_). Special characters (e.g., @, #, \$) and spaces are **not** allowed.

2. Getting Started

Prerequisites

- **Java SE 21** must be installed on your machine.

Launching the Application CLEVIS is a command-line application. You must launch it from your terminal or command prompt. When launching, you **must** specify the paths for the log files using `-html` and `-txt` flags.

Command Syntax: `java hk.edu.polyu.comp.comp2021.clevis.Application -html <html_log_path> -txt <txt_log_path>`

Example: `java hk.edu.polyu.comp.comp2021.clevis.Application -html log.html -txt log.txt`

Successful Launch Output:

HTML logger: log.html

TXT logger: log.txt

Clevis - Command Line Vector Graphics Software
clevis>

Once you see the clevis> prompt, the system is ready for your commands.

3. Commands Description

3.1 Shape Creation

All coordinate values (x, y, w, h, etc.) are floating-point numbers (e.g., 10.5, 20).

Shape Type	Command Syntax	Description
Rectangle	rectangle <n> <x> <y> <w> <h>	Creates a rectangle named n at top-left (x,y) with width w and height h.
Square	square <n> <x> <y> <l>	Creates a square named n at top-left (x,y) with side length l.
Circle	circle <n> <x> <y> <r>	Creates a circle named n centered at (x,y) with radius r.
Line	line <n> <x1> <y1> <x2> <y2>	Creates a line named n from point (x1,y1) to (x2,y2).

Visual Examples:

```
Clevis - Command Line Vector Graphics Software
clevis> rectangle r1 10 20 20 10
clevis> Rectangle 'r1' created successfully.
clevis> square s1 10 20 10
clevis> Square 's1' created successfully.
clevis> circle c1 5 10 5
clevis> Circle 'c1' created successfully.

clevis> line l1 15 30 23 12
clevis> Line 'l1' created successfully.
```

Figure 1-2: Four different shapes (Rectangle, Square, Circle and Line) are created.

3.2 Grouping & Manipulation

Command	Syntax	Description
Group	group <name> <n1> <n2> ...	Groups existing shapes n1, n2... into a new group named name. Grouped shapes are locked and cannot be selected individually.
Ungroup	ungroup <name>	Dissolves the group name, restoring the individual members and their original layer order.
Move	move <n> <dx> <dy>	Moves shape or group n by dx (horizontal) and dy (vertical).
Delete	delete <n>	Permanently removes shape n from the canvas.

Note: You cannot move or delete a shape that is currently inside a group. You must ungroup it first.

Visual Examples:

```
Clevis - Command Line Vector Graphics Software
clevis> square s1 0 0 4
clevis> Square 's1' created successfully.
clevis> circle c1 0 0 1
clevis> Circle 'c1' created successfully.
clevis> rectangle r1 0 0 1 2
clevis> Rectangle 'r1' created successfully.
clevis> line l1 0 0 1 1
clevis> Line 'l1' created successfully.
```

```

clevis> group g1 s1 c1 r1 l1
clevis> Group 'g1' created successfully.
clevis> move g1 1 1
clevis> Shape 'g1' moved by (1.0, 1.0) successfully.
clevis> delete g1
clevis> Shape 'g1' deleted successfully.

```

Figure 3-4: Demonstration of group, move and delete

```

clevis> square s2 0 0 4
clevis> Square 's2' created successfully.
clevis> circle c2 0 0 1
clevis> Circle 'c2' created successfully.
clevis> rectangle r3 0 0 1 2
clevis> Rectangle 'r3' created successfully.
clevis> group g2 s2 c2 r3
clevis> Group 'g2' created successfully.
clevis> ungroup g2
clevis> Group 'g2' ungrouped successfully.

```

Figure 3-4: Demonstration of group and ungroup

3.3 Analysis & Queries

Command	Syntax	Description
Bounding Box	boundingbox <n>	Displays the smallest upright rectangle that encloses shape n.
Shape At	shapeat <x> <y>	Identifies the topmost shape (highest Z-order) covering the point (x,y).

Intersect	intersect <n1> <n2>	Checks if the bounding boxes of n1 and n2 overlap.
List	list <n>	Shows detailed information (coordinates, dimensions) of shape n.
List All	listall	Lists all shapes on the canvas in decreasing Z-order . Uses indentation to show group hierarchy.

Visual Examples:

```

HTML logger:log.html
TXT logger:log.txt
Clevis - Command Line Vector Graphics Software
clevis> square S1 0 0 4
clevis> Square 'S1' created successfully.
clevis> circle C1 0 0 5
clevis> Circle 'C1' created successfully.
clevis> line L1 3 3 4 4
clevis> Line 'L1' created successfully.
clevis> group G1 S1 C1
clevis> Group 'G1' created successfully.
clevis> listall
Group: name = G1, z = 3, members = 2
  - S1 (Square)
  - C1 (Circle)
Line: name = L1, from = (3.00, 3.00), to = (4.00, 4.00)
clevis> shapeat 0 0
clevis> The shape at (0.0, 0.0) is 'G1'.
clevis> intersect G1 L1
clevis> 'G1' and 'L1' intersect.
clevis> boundingbox G1
Bounding box: top left = (-5.00, -5.00), width = 10.00, height = 10.00
clevis> list L1
Line: name = L1, from = (3.00, 3.00), to = (4.00, 4.00)
clevis> 

```

Figure 5: The usage of these commanders (bounding box, shape at, intersect, list and list all, including the special case for group)

3.4 Project History

CLEVIS supports full Undo/Redo functionality for all modification commands (Create, Move, Delete, Group, Ungroup).

CLEVIS supports comprehensive logging functionality that automatically records all modification commands(Create, Move, Delete, Group, Ungroup, Undo, Redo).

Command	Syntax	Description
Undo	undo	Reverts the last modification.

Redo	redo	Re-applies the previously undone action.
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3.5 System Commands

Command	Syntax	Description
Show help	help	show help and show all the syntax
Quit	quit	Saves logs, cleans up resources, and exits the application.

4. Troubleshooting

Below are common error messages and how to resolve them.

Issue 1: "Shape name ... already exists"

- **Cause:** You are trying to create a shape with a name that is already in use.
- **Solution:** Use listall to see current names, or delete the old shape first. Shape names must be unique.

Issue 2: "Shape is locked in a Group"

- **Cause:** You tried to move or delete a shape (e.g., EyeL) that is currently part of a Group (e.g., Face).
- **Solution:** You must ungroup Face first if you want to modify EyeL individually.

Issue 3: "Dimensions must be positive"

- **Cause:** You entered a negative number or zero for width, height, or radius.
- **Solution:** Re-enter the command with positive values (e.g., radius must be > 0).

Issue 4: Logs are not generated

- **Cause:** The program might have been terminated forcefully (e.g., closing the window/Ctrl+C) instead of using the command.
- **Solution:** Always use the quit command to ensure logs are saved and closed properly.

Issue 5: "Error: '...' is not a valid number."

- **Cause:** You entered non-numeric characters (e.g., text like "abc" or symbols) for a coordinate or dimension. The system detects this and prevents the crash.

- **Solution:** Re-enter the command using valid numerical values (e.g., 10, 5.5, -20).

Issue 6: "Error: Invalid name... Names must contain only letters, numbers, or underscores."

- **Cause:** You attempted to create a shape or group using a name containing restricted characters (such as @, #, \$, or spaces).
- **Solution:** Rename the shape using only letters, numbers, and underscores (e.g., use rect_1 instead of rect#1).

Issue 7: "Error: Empty command."

- **Cause:** You pressed Enter without typing a command, or entered a string of only spaces.
- **Solution:** Type a valid command (e.g., listall) and press Enter.